

On direct multilevel solvers induced by AMG

Michal Outrata, José Pablo Lucero Lorca,
Conor McCoid

Set-up



Set-up

$$L\mathbf{u} = \mathbf{f}$$

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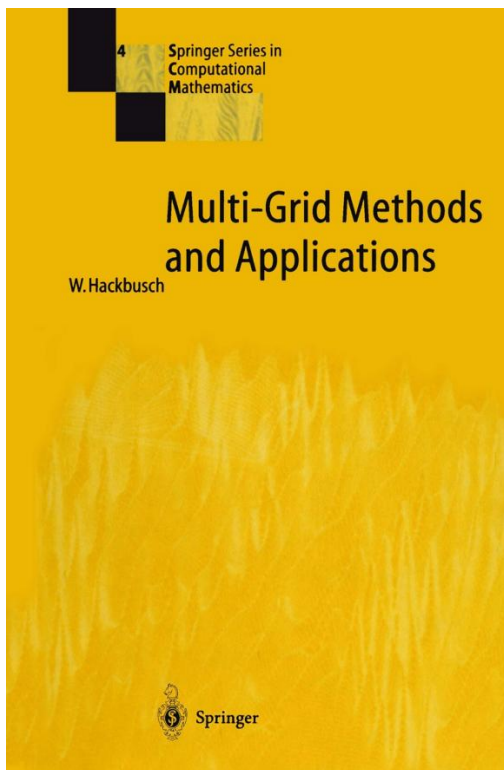
$$L\mathbf{u} = \mathbf{f}$$

$$L \in \mathbb{C}^{N \times N} \quad \& \quad \mathbf{f} \in \mathbb{C}^N$$

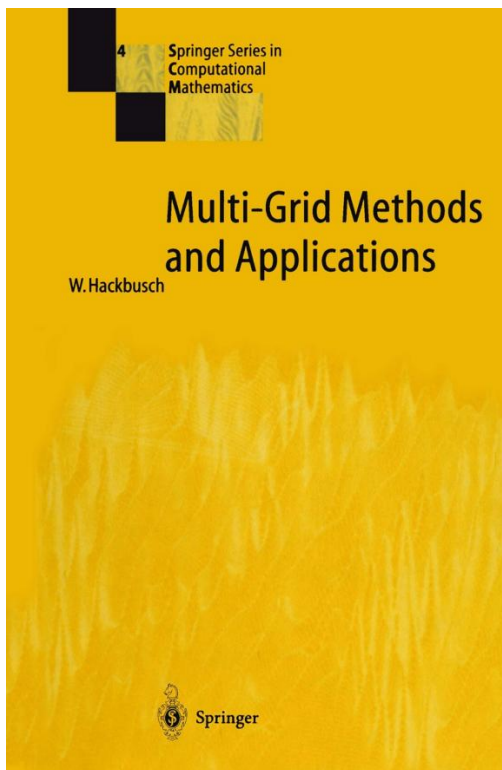
Multigrid tutorial



Multigrid tutorial



Multigrid tutorial



A natural model of an elliptic equation is the two-dimensional Poisson equation in a square. Unfortunately, the analysis of the multi-grid algorithm for this problem is still an involved exercise. Therefore, we start with the simplest *one*-dimensional Dirichlet boundary value problem

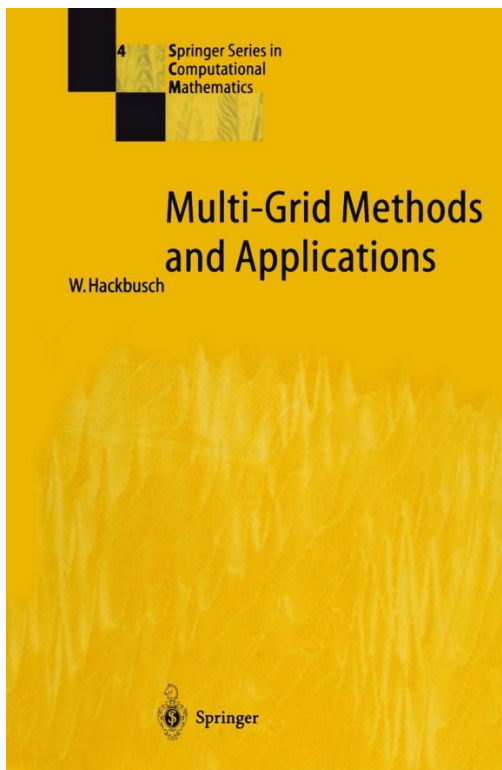
$$-u''(x) = f(x) \quad \text{in} \quad \Omega = (0, 1), \quad u(x) = 0 \quad \text{at} \quad x \in \Gamma = \{0, 1\}. \quad (2.1.1)$$

Multigrid tutorial



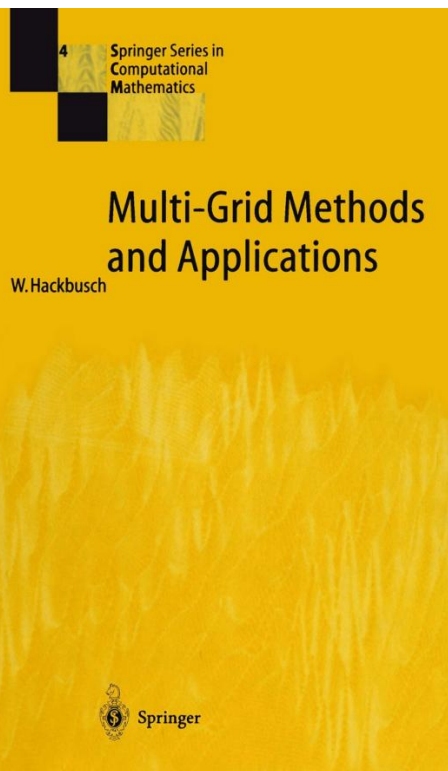
Based on a tutorial from Scott MacLachlan at Memorial University, Newfoundland.

Multigrid tutorial



The characteristic feature of the multi-grid iteration is its fast convergence. The convergence speed does not deteriorate when the discretisation is refined, whereas classical iterative methods slow down for decreasing grid size. As a consequence one obtains an acceptable approximation of the discrete problem at the expense of computational work proportional to the number of unknowns, which is also the number of the equations in the system. It is not only the complexity which is optimal, also the constant of proportionality is so small that other methods can hardly surpass the multi-grid efficiency.

Multigrid tutorial



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The previous characterisation does not mean that there is a fixed multi-grid algorithm applying to all boundary value problems. There is rather a multi-grid technique fixing only the framework of the algorithm. The efficiency of the multi-grid algorithm depends on the adjustment of its components to the problem in question. Therefore it is one task of this book to describe the basic framework and the scope of several of the different multi-grid components.

Multigrid framework



Multigrid framework

Algorithm 1 Two-level AMG $\mathbf{v} \mapsto M^{-1}\mathbf{v}$

1:

2:

3:

4:

5:

6:

7:

8:

9:

Multigrid framework

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```

pre – smoothing
 $\alpha_1, \dots, \alpha_m, S^{-1}$

Multigrid framework

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$\alpha_1, \dots, \alpha_m, S^{-1}$

coarse correction

P, R, M_0^{-1}

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AMG background



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$$L = \begin{bmatrix} L_{ff} & L_{fc} \\ L_{cf} & L_{cc} \end{bmatrix} \equiv \begin{bmatrix} A & B \\ C & D \end{bmatrix} \in \mathbb{C}^{N \times N}$$

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$$L_{ff} \equiv A \in \mathbb{C}^{(N-n) \times (N-n)}$$

AMG background

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$$L_{ff} \equiv A \in \mathbb{C}^{(N-n) \times (N-n)}$$

$$L_{cc} \equiv D \in \mathbb{C}^{n \times n}$$

Existing direct method



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LDU factorization of L^{-1}

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$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}^{-1} = \begin{bmatrix} I & -A^{-1}B \\ & I \end{bmatrix} \begin{bmatrix} A^{-1} & \\ & K^{-1} \end{bmatrix} \begin{bmatrix} I & \\ -CA^{-1} & I \end{bmatrix}$$

Existing direct method

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$$K = D - CA^{-1}B$$

Existing direct method

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$$m = 1 \quad \alpha_1 = 1$$

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$m = 1$ $\alpha_1 = 1$ no post – smoothing (lines 6 – 8)

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$$S^{-1} = \begin{bmatrix} A^{-1} & \\ & \end{bmatrix} \quad P = \begin{bmatrix} -A^{-1}B \\ I \end{bmatrix} \quad R = \begin{bmatrix} \star & I \end{bmatrix} \quad M_0^{-1} = (RLP)^{-1} = K$$

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T. A. Manteuffel, S. Münzenmaier,
J. Ruge, and B. Southworth.

Nonsymmetric reduction-based
algebraic multigrid.

Journal on Scientific Computing,
2019.

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Why direct methods?



New question



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symmetric/palindromic structure of the cycle

Resulting method



Resulting method

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direct?

Analysis



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Error propagation:

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Error propagation:

$$L^{-1}\mathbf{v} - M^{-1}\mathbf{v} = ?$$

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Error propagation

$$E^{(s)}(\alpha_i) = I - \alpha_i S^{-1}L$$

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$$E^{(c)} = I - PM_0^{-1}RL$$

Analysis

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Error propagation

$$E(\alpha_1, \dots, \alpha_m) = \left(\prod_{i=m}^1 E^{(s)}(\alpha_i) \right) E^{(c)} \left(\prod_{i=1}^m E^{(s)}(\alpha_i) \right)$$

Spectral analysis



Spectral analysis

$$T := A^{-1}BD^{-1}C$$

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Assumptions

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$$E^{(s)}\tilde{V} = \begin{bmatrix} (1-\alpha)I & -\alpha A^{-1}B \\ -\alpha D^{-1}C & (1-\alpha)I \end{bmatrix} \begin{bmatrix} W\sqrt{\Lambda} & W\sqrt{\Lambda} \\ D^{-1}CW & -D^{-1}CW \end{bmatrix}$$

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$$E^{(s)} \sim \begin{bmatrix} (1 - \alpha - \alpha\sqrt{\Lambda})I & \\ & (1 - \alpha + \alpha\sqrt{\Lambda})I \end{bmatrix}$$

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$$E^{(c)}\tilde{V} = \begin{bmatrix} W\sqrt{\Lambda} & W\sqrt{\Lambda} \\ D^{-1}CW & -D^{-1}CW \end{bmatrix} \begin{bmatrix} \frac{1}{2}(I + \sqrt{\Lambda}) & \frac{1}{2}(I - \sqrt{\Lambda}) \\ \frac{1}{2}(I + \sqrt{\Lambda}) & \frac{1}{2}(I - \sqrt{\Lambda}) \end{bmatrix}$$

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Spectral analysis

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eigenproperties of $E(\alpha_1, \dots, \alpha_m)$ given by matrices in $\mathbb{C}^{2 \times 2}$

Spectral analysis

$$T := A^{-1}BD^{-1}C$$

Assumptions

T is diagonalizable

Spectral analysis

Proposition. *If $(\lambda, \mathbf{w})$ is an eigenpair of T with $\lambda \neq 0$, then*

Spectral analysis

Proposition. *If $(\lambda, \mathbf{w})$ is an eigenpair of T with $\lambda \neq 0$, then*

$$\left(0, \begin{bmatrix} \left(\rho_+(1 - \sqrt{\lambda}) - \rho_-(1 + \sqrt{\lambda}) \right) \sqrt{\lambda} \mathbf{w} \\ \left(\rho_+(1 - \sqrt{\lambda}) + \rho_-(1 + \sqrt{\lambda}) \right) D^{-1} C \mathbf{w} \end{bmatrix} \right), \quad \left(\rho_\lambda, \frac{1}{2} \begin{bmatrix} (\rho_- + \rho_+) \sqrt{\lambda} \mathbf{w} \\ (\rho_- - \rho_+) D^{-1} C \mathbf{w} \end{bmatrix} \right)$$

are eigenpairs of $E(\alpha_1, \dots, \alpha_m)$ with

$$\rho_\lambda := \frac{1}{2} \left((1 + \sqrt{\lambda}) \rho_-^2 + (1 - \sqrt{\lambda}) \rho_+^2 \right) \quad \text{and} \quad \rho_\pm := \prod_{i=1}^m (1 - \alpha_i \pm \alpha_i \sqrt{\lambda}).$$

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$$\left(0, \begin{bmatrix} 0 \\ D^{-1}C\mathbf{z} \end{bmatrix}\right), \quad \left(\prod_{i=1}^m (1 - \alpha_i)^2, \begin{bmatrix} \prod_{i=1}^m (1 - \alpha_i) \mathbf{z} \\ \sum_{i=1}^m (-\alpha_i) \prod_{j \neq i} (1 - \alpha_j) D^{-1}C\mathbf{z} \end{bmatrix}\right),$$

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$$\{0\} \cup \{\rho_\lambda\}_{\lambda \in \sigma(T)}$$

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Direct ?



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$$m = 1: \quad \rho_\lambda(\alpha) = (1 - \alpha)^2 + \alpha(3\alpha - 2)\lambda$$

Direct ?

Proposition. *Let $m \geq 1$ and for $i = 1, \dots, m$ set*

$$c_i = -\cos\left(\frac{2\pi i}{2m+1}\right) \quad \text{and} \quad \hat{\alpha}_i := \frac{1}{1+c_i}.$$

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Hence, the spectrum of $E(\alpha_1, \dots, \alpha_m)$ is clustered in two points,

$$\sigma(E(\alpha_1, \dots, \alpha_m)) = \left\{ 0, \frac{1}{(2m+1)^2} \right\}.$$

The direct method



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$$(M^{-1}L - I) \left(M^{-1}L - \frac{(2m+1)^2 - 1}{(2m+1)^2} I \right) = 0$$

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two applications $\longrightarrow$ direct

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two applications $\longrightarrow$ direct

two iterations of CG/GMRES $\longrightarrow$ direct

The multivelel direct method



The multivelel direct method

Algorithm 1 Two-level AMG $\mathbf{v} \mapsto M^{-1}\mathbf{v}$

```
1:  $\mathbf{x} \leftarrow \mathbf{0}$ 
2: for  $i = 1, \dots, m$  do
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W – cycle

The multivelel direct method

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W – cycle

K – cycle

Conclusion



**Thank you for
your attention**

Numerical example



Numerical example



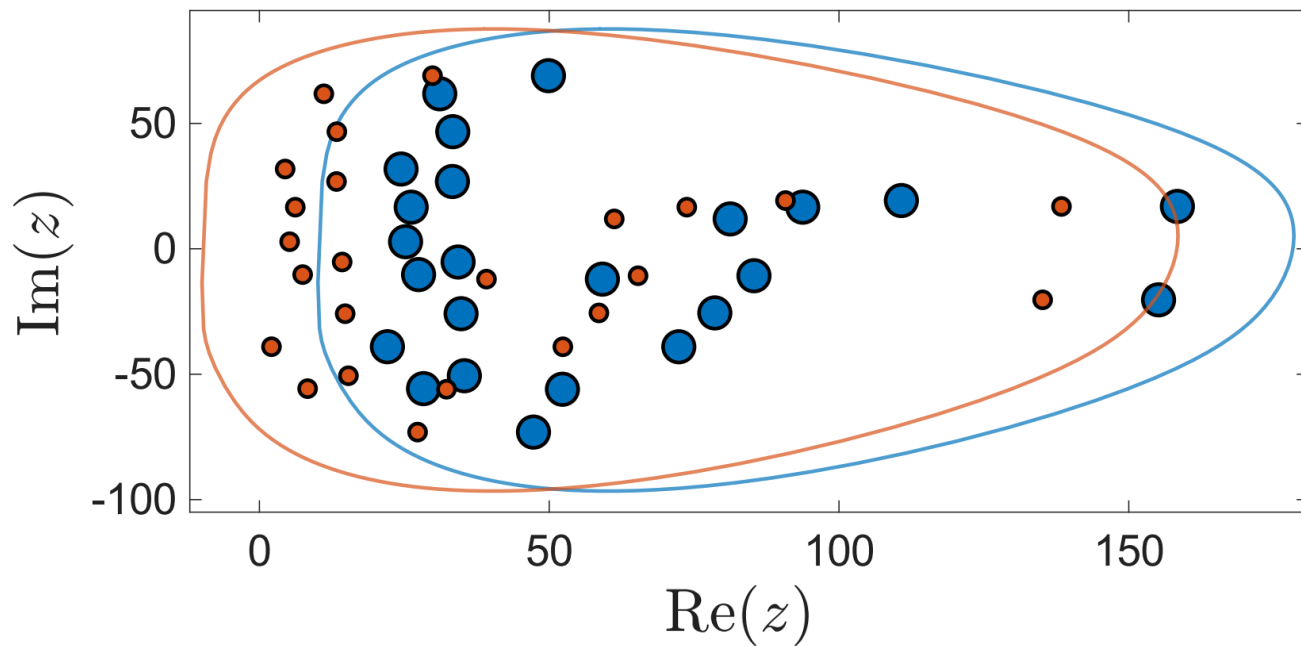
<https://pablo.world/CCIMM26/#/5>

Numerical example

$$L = H + \gamma K$$

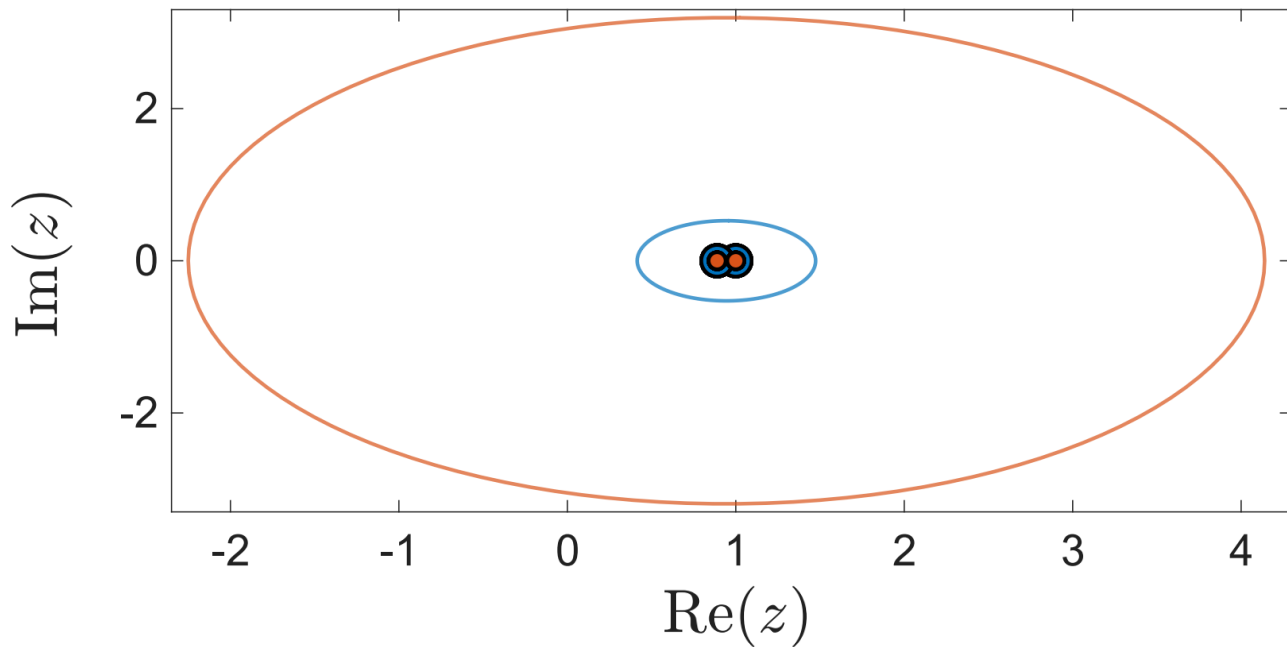
Numerical example

$L, N = 24 \quad \kappa \approx 9.9 \quad \& \quad 52$



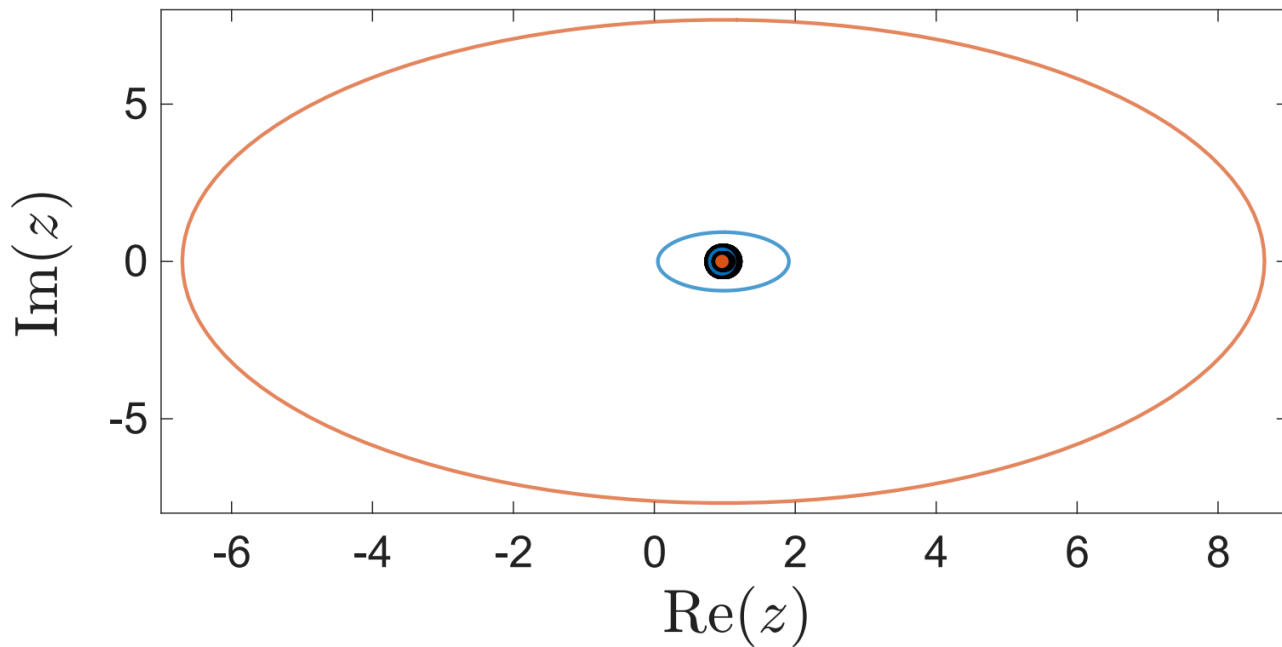
Numerical example

$$M^{-1}L, m = 1 \quad \kappa \approx 2.9 \quad \& \quad 48$$



Numerical example

$$M^{-1}L, m = 2 \quad \kappa \approx 5.4 \quad \& \quad 2.5\text{e}+02$$



Numerical example

